O-D-210A
February 25, 1976
SUPERSEDING
Fed. Spec. O-D-210
July 14, 1959

FEDERAL SPECIFICATION

DESICCANT, ACTIVATED, CALCIUM SULFATE, ANHYDROUS

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers two types of anhydrous calcium sulfate for use as a desiccant.

1.2 Classification. The desiccant shall be of the following types and classes as specified (see 6.2):

Type I - Non-indicating.

Class 1 - Nominal 8 mesh.
Class 2 - Nominal 6 mesh.

Type II - Indicating (nominal 8 mesh).

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issues in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein:

Federal Specifications:

QQ-A-250/4	-	Aluminum Alloy 2024, Plate and Sheet.
QQ-M-44	-	Magnesium Alloy Plate and Sheet (AZ31B).
QQ-S-698	-	Steel, Sheet and Strip, Low-Carbon.
RR-S-366	-	Sieve, Test.
PPP-B-591	-	Boxes, Shipping, Fiberboard, Wood-Cleated.
PPP-B-601	-	Boxes, Wood, Cleated-Plywood.
PPP-B-621	-	Boxes, Wood, Nailed and Lock-Corner.
PPP-B-636	-	Boxes, Shipping, Fiberboard.
PPP-C-96	-	Cans, Metal, 28 Gage and Lighter.
PPP-T-60	-	Tape: Packaging, Waterproof.

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Federal Standards:

Fed. Std. No. 123 - Marking for Shipment (Civil Agencies).
Fed. Std. No. 356 - Commercial Packaging of Supplies and Equipment.

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U. S. Government Printing Office, Washington, DC 20402.

(Single copies of this specification and other Federal specifications required by activities outside the Federal Government for bidding purposes are available without charge from Business Service Centers at the General Services Administration Regional Offices in Boston, New York, Washington, DC, Atlanta, Chicago, Kansas City, MO, Fort Worth, Denver, San Francisco, Los Angeles, and Seattle, WA.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Military Standards:

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.

(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on date of invitation for bids or request for proposal shall apply.

Uniform Classification Committee:

Uniform Freight Classification.

(Application for copies should be addressed to the Uniform Classification Committee, Room 1106, 222 South Riverside Plaza, Chicago, IL 60606.)

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National Classification Board:

National Motor Freight Classification.

(Application for copies should be addressed to the American Trucking Associations, Inc., Tariff Order Section, 1616 P Street, N.W., Washington, DC 20036.)

American Society for Testing and Materials (ASTM) Standards:

C 471-72 - Chemical Analysis of Gypsum and Gypsum Products.
D 1193-74 - Reagent Water.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

American National Standards Institute (ANSI) Standard:

B46.1 - Surface Texture, Surface Roughness, Waviness and Lay.

(Application for copies should be addressed to the American National Standards Institute, 1430 Broadway, New York, NY 10018.)

3. REQUIREMENTS

3.1 Material. The material shall be anhydrous calcium sulfate which is free-flowing and noncaking when tested as specified in 4.2.4.1.

3.2 Purity. Type I material shall contain no less than 38.5 percent by weight calcium oxide (CaO) and no less than 56.0 percent by weight sulfur trioxide (SO_3) when tested as specified in 4.2.4.2. Type II material shall contain no less than 37.0 percent by weight CaO and no less than 54.5 percent by weight SO_3 when tested as specified in 4.2.4.2.

3.3 Water absorption. Type I and type II material shall absorb no less than 9.0 percent by weight water when tested as specified in 4.2.4.3.

3.4 Indicator. Type II material shall be impregnated with no less than 3.0 and no more than 5.0 percent by weight anhydrous cobaltous chloride ($CoCl_2$) when tested as specified in 4.2.4.4. The type II material shall be blue in color and shall change to pink or rose when tested for water absorption as specified in 4.2.4.3.

3.5. Corrosiveness. When specified in the contract or order (see 6.2), type I material shall be noncorrosive when tested as specified in 4.2.4.5.

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3.6 Particle size characteristics. Type I, class 1 material and type II material shall conform to the particle size characteristics of table I when tested as specified in 4.2.4.6. Type I, class 2 material shall conform to the particle size characteristics of table II when tested as specified in 4.2.4.6.

TABLE I. Particle size of type I, class 1 and type II material

Characteristic	Percent by weight
Retained on 4,760-micron (No. 4) sieve, maximum	1
Passing 4,760-micron (No. 4) sieve and retained on 3,360-micron (No. 6) sieve, maximum	25
Passing 3,360-micron (No. 6) sieve and retained on 1,410-micron (No. 14) sieve, minimum	72
Passing 590-micron (No. 30) sieve, maximum	2

TABLE II. Particle size of type I, class 2 material

Characteristic	Percent by weight
Retained on 5,660-micron (No. 3-1/2) sieve, maximum	10
Passing 5,660-micron (No. 3-1/2) sieve and retained on 3,360-micron (No. 6) sieve, minimum	90
Passing 590-micron (No. 30) sieve, maximum	2

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any

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other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to specified requirements.

4.2 Quality conformance inspection.

4.2.1 Lotting. A lot shall consist of the desiccant of one type and class produced by one manufacturer, at one plant, from the same materials, and under essentially the same manufacturing conditions provided the operation is continuous. In the event the process is a batch operation, each batch shall constitute a lot (see 6.3).

4.2.2 Sampling.

4.2.2.1 For examination of preparation for delivery. Sampling for examination of preparation for delivery shall be conducted in accordance with MIL-STD-105, inspection level S-2.

4.2.2.2 For test (see 6.4). Sampling for test shall be conducted in accordance with table III. A representative specimen of approximately 250 grams (g) shall be removed from each sample container and placed in a suitable clean, dry container labeled to identify the lot and container from which it was taken.

TABLE III. Sampling for test

Number of containers in batch or lot	Number of sample containers
2 to 25	2
26 to 150	3
over 151	5

4.2.3 Inspection procedure.

4.2.3.1 For examination of preparation for delivery. The lot shall be the number of containers offered for delivery at one time. The sample unit shall be one filled container, ready for shipment. The containers shall be examined for the following defects using inspection level S-2 and an AQL of 4.0 percent defective:

(a) Contents per container not as specified
(b) Container not as specified

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(c) Container closure not as specified
(d) Container damaged or leaking
(e) Tape missing or not as specified (where required)
(f) Marking incorrect, missing, or illegible

4.2.3.1.1 Examination of closure, water-proofing, banding or strapping of shipping containers. When shipping containers are required to comply with PPP-B-591, PPP-B-601, PPP-B-621, or PPP-B-636, examination for defects in closure, waterproofing, banding, or strapping shall be in accordance with the appendix of that specification.

4.2.3.2 For test. Each sample specimen taken in 4.2.2.2 shall be tested as specified in 4.2.4. Failure of any test by any specimen shall be cause for rejection of the lot represented.

4.2.4 Tests (see 6.4). Water in accordance with ASTM D 1193-74 and reagent grade chemicals shall be used throughout the tests. Where applicable, blank determinations shall be run and corrections applied where significant. Tests shall be conducted as follows:

4.2.4.1 Material. Examine the specimen for flowing properties and evidence of caking.

4.2.4.2 Purity. Determine the percent by weight CaO and SO3 in the specimen in accordance with the applicable portions of ASTM C 471-72.

4.2.4.3 Water absorption. Weigh to the nearest milligram (mg) approximately 5.0 g of the specimen and spread uniformly over the bottom of a tared moisture dish approximately 55 millimeters in diameter. Place the open dish over water in a desiccater and allow to stand for 24 hours at 20° to 30° C. If type II material is being tested, observe the color of the hydrated specimen. Reweigh the dish and specimen to the nearest milligram and calculate the percent by weight water absorption as follows:

$$\text{Percent water absorption} = \frac{100(A - B)}{W}$$

where: A = Weight of specimen and dish after water absorption in grams,
B = Weight of specimen and dish before water absorption in grams, and
W = Weight of specimen in grams.

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4.2.4.4 Indicator. Weigh to the nearest 10 mg approximately 10 g of the specimen (type II material only) and transfer to a beaker with 200 milliliters (ml) of chloride-free water. Agitate gently to dissolve the soluble portion ($CoCl_2$) and transfer, with or without filtration, to a 500-ml volumetric flask and dilute to the volume mark with chloride-free water. Mix thoroughly, allow to settle if not filtered, and withdraw a 100-ml aliquot of the soluiton. Titrate the aliquot in a beaker with 0.1N silver nitrate solution using potassium chromate solution as an indicator.

Calculate the percent by weight indicator as $CoCl_2$ as follows:

$$\text{Percent } CoCl_2 = \frac{6.493AB}{W}$$

where: A = Milliliters of silver nitrate solution used in the titration,
B = Normality of the silver nitrate solution, and
W = Weight of specimen in aliquot in grams.

4.2.4.5 Corrosiveness. This test shall be conducted on type I material when the requirement for corrosiveness is specified in the contract or order.

(a) Preparation of specimen. Grind approximately 20 g of the specimen to a fineness of approximately 200 to 400 microns (40 to 80 mesh). Assemble two sieves, one 420 microns (No. 40) and one 177 microns (No. 80) conforming to RR-S-366. Thoroughly sift the ground specimen through the No. 40 sieve onto the No. 80. Place the portion of the specimen which passes through the No. 40 and is retained on the No. 80 sieve in a desiccator maintained at 20° ± 1° C and 60 percent relative humidity by means of a 38.4 percent by weight sulfuric acid in water solution. Condition the specimen in the desiccator for at least 24 hours.

(b) Preparation of test panels. Prepare a 2 inch by 1 inch by 1/8 inch test panel of each of the following materials: steel conforming to hot rolled, commercial quality, sheet or strip of QQ-S-698; magnesium alloy conforming to temper condition O of QQ-M-44; and aluminum alloy conforming to temper of O of QQ-A-250/4. Grind and polish the surface of each test panel to a 6 to 8 root mean square average microinch roughness as defined by ANSI B46.1. Clean each test panel by 2 minutes immersion in toluene at 100° ± 1° C followed by a quick rinse in acetone at 50° ± 1° C and then by a quick rinse in methanol at 50° ± 1° C. During and after cleaning, handle the panels with forceps and on the edge only, or by wire loops passing through previously drilled holes near one corner. After the methanol rinse, dry each panel for 1 minute, but no more than 2 minutes, in an oven at 38° ± 1° C.

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(c) Procedure. Cover approximately one-half of each test panel prepared in (b) with the conditioned specimen prepared in (a). Place the specimens thus prepared in a desiccator maintained at 60 percent relative humidity with sulfuric acid solution having a specific gravity of 1.30 to 1.31 at 20°/20°C. Place the desiccator in an oven at 38° ± 1° C for 72 hours. At the end of the 72 hours, remove the test panels, brush away the specimen with a camel's hair brush, and visually compare the portions of the test panel that were covered and uncovered. If any of the portions that were covered are corroded more than the corresponding portions that were uncovered, the specimen shall be considered to be corrosive.

4.2.4.6 Particle size characteristics. Use sieves conforming to RR-S-366. Assemble the appropriate size sieves specified in 3.6 with a bottom pan and place in a motor-driven mechanical sieve shaker. Place approximately 100 g of specimen on the top sieve and shake for 5 minutes. Weigh the portions retained on and passing each sieve and calculate the percent retained and passing as required.

5. PREPARATION FOR DELIVERY

5.1 Packaging. The desiccant shall be packaged level A or C as specified (see 6.2).

5.1.1 Level A.

5.1.1.1 Unit packaging. The desiccant packaged in 200-g, 1-pound, 2-pound, or 5-pound (+1 or -0 percent) quantities as specified (see 6.2) shall be packaged within containers conforming to type V, class 4, round or square, with 1-3/4-inch metal screw-cap closure of PPP-C-96 or within containers conforming to type VIII, round or cone top, with 1-3/4-inch screw-cap closure of PPP-C-96. The containers shall have inner and outer seals and closures torque-closed to 28 to 32 pound-inches.

5.1.1.2 Intermediate packaging.

(a) Two hundred g, 1-pound or 2-pound quantities. Twenty-four 200-g, twelve 1-pound, or twelve 2-pound quantities of desiccant packaged as specified in 5.1.1.1 shall be intermediately packaged in a snug-fitting fiberboard box conforming to type CF, class weather-resistant, variety SW, grade W5c or W6c of PPP-B-636. Twenty-four of the 200-g quantity containers shall be placed in the box in two layers of three rows of four unit containers per row while 12 of the 1-pound or 2-pound quantities shall be placed in the same manner but in only one layer. The fiberboard boxes shall be closed in accordance with the closure requirements

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specified in the appendix to PPP-B-636. All box seams, including the manufacturer's joint, shall be taped with minimum 3-inch wide pressure-sensitive tape conforming to type III, class 1, color optional of PPP-T-60. The tape shall be extended over all corners and edges at least 3 inches.

(b) Five-pound quantity. The desiccant packaged as specified in 5.1.1.1 shall be intermediately packaged as specified in (a) except that six containers shall be placed in a fiber box in one layer of two rows with three containers in each row.

5.1.2 Level C.

5.1.2.1 Civil agencies. The desiccant shall be packaged to afford adequate protection against damage during shipment from the supply source to the first receiving agency.

5.1.2.2 Military activities. The desiccant shall be packaged in accordance with Fed. Std. No. 356.

5.2 Packing. The desiccant shall be packed level A, B, or C as specified (see 6.2).

5.2.1 Level A. The desiccant packaged as specified in 5.1 shall be packed not to exceed container weight limitations in snug-fitting type overseas containers conforming to PPP-B-591, PPP-B-601, or PPP-B-621. The containers shall be closed and strapped in accordance with their respective specification requirements using galvanized strapping.

5.2.2 Level B. The desiccant packaged as specified in 5.1 shall be packed not to exceed container weight limitations in snug-fitting type domestic containers conforming to PPP-B-591, PPP-B-601, or PPP-B-621 or in a fiberboard box conforming to type CF, class weather-resistant, variety SW, grade W5c of PPP-B-636. The containers shall be closed, waterproofed, and banded or strapped in accordance with the requirements specified in the appendix to the applicable box specification.

5.2.3 Level C.

5.2.3.1 Civil agencies. Desiccant in quantities as specified (see 6.2) shall be packed in a container complying with the Uniform or National Motor Freight Classification, as applicable. The containers shall be packed to assure carrier acceptance and safe arrival at destination.

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5.2.3.2 Military activities. The desiccant shall be packed in accordance with Fed. Std. No. 356.

5.3 Marking. Marking shall include the manufacturer's recommendations for use and reactivation.

5.3.1 Military activities. Unit, intermediate, and shipping containers shall be marked in accordance with MIL-STD-129 for level A and level B containers and in accordance with Fed. Std. No. 356 for level C containers. All containers shall be marked to show the lot number and date of manufacture.

5.3.2 Civil agencies. Unit, intermediate, and shipping containers shall be marked in accordance with Fed. Std. No. 123.

6. NOTES

6.1 Intended use. Type I, class 1 material is intended for use in laboratory units for drying air and other gases. Type I, class 2 material is intended for general use such as in large drying units and drying liquids either in liquid or vapor phase. Type II material is intended for use in maintaining dewpoints at or below -65°F in a closed system.

6.2 Ordering data. Purchasers should select the preferred options permitted herein and include the following information in procurement documents:

(a) Title, number, and date of this specification.
(b) Type and class of desiccant required (see 1.2).
(c) If type I material shall meet corrosiveness requirement (see 3.5).
(d) Level of packaging and packing required (see 5.1 and 5.2).
(e) Unit quantity required (see 5.1.1.1).

6.3 Batch. A batch is defined as that quantity of material which has been manufactured by some unit chemical process or subjected to some physical mixing operation intended to make the final product substantially uniform.

6.4 Sampling and testing precautions. Anhydrous calcium sulfate gains weight rapidly when exposed to a moist atmosphere and every precaution should be taken to minimize this exposure during sampling and testing.

6.5 Significant places. For the purpose of determining conformance with this specification, an observed or calculated value shall be

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rounded off "to the nearest unit" in the last right-hand place of figures used in expressing the limiting value, in accordance with the rounding-off method of the Recommended Practice for Indicating Which Places of Figures Are To Be Considered Significant in Specified Limiting Values (ASTM E 29).

MILITARY INTERESTS:

Custodians:

Army - EA
Air Force - 68

Review activities:

Army - SM
DSA - GS

Civil Agency Coordinating Activities:

GSA - FSS
VA - DMS

Preparing activity:

Army - EA

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